



AI CODING FOR VFX ARTISTS

Setup Guide

Install your AI assistant and the basics (ffmpeg, Python, GitHub) on Windows, macOS or Linux. Mostly by letting the assistant do it for you.

John Huikku · Checked September 2026 · app menus and labels may differ slightly between versions

How this guide works

You only do step 1 by hand. After that, you paste a **setup prompt** and the assistant installs things for you, asking before each step.

Every step also has **manual commands** in case you'd rather do it yourself, or the assistant gets stuck.

Step	What	Needed for
1	Install your AI tool and sign in	Everything
2	Get the class folder	Everything
3	Check your machine (read-only)	Knowing what's missing
4	ffmpeg	Converting video, contact sheets
5	Python + uv	Small tools and scripts
6	Git + GitHub	Undo button, sharing tools
7	Starter pack for your own projects	Studio rules in your real work
8	Optional: MCP for Blender, Node.js, Tauri	Driving Blender, web apps, desktop apps

The setup prompts are in `starter-pack/prompts/setup-prompts.md` and printed at the back of this guide. **Copy them from the .md file**, because PDFs can mangle quotes.

Approving installs. On Windows you may see a "Do you want to allow this app to make changes?" box. On a Mac you may be asked for your password. That's normal for installing software. If your computer is managed by IT and blocks installs, ask IT; don't try to work around it.

1. Install your AI tool

Pick **one**: the one your subscription includes. No subscription? Codex (ChatGPT Free) and Antigravity (free tier) both work without paying.

Claude Code

1. Download the **Claude desktop app** from claude.com/download (Windows and Mac; Linux is in beta).
2. Sign in. Claude Code needs a **Pro, Max or Team** plan.
3. Open the **Code** tab and choose your working folder (step 2).

Codex

1. Download the **ChatGPT desktop app** from chatgpt.com/download (Windows, Mac, Linux).
2. Sign in with your ChatGPT account (any plan, including Free).
3. Switch to **Codex** mode and choose your working folder.

Antigravity

1. Download **Antigravity** from antigravity.google/download (Windows, Mac, Linux).
2. Sign in with your Google account (free tier, or higher limits with Google AI Pro).
3. Open your working folder as a workspace.

Editor and terminal versions (VS Code extension, Antigravity IDE, the CLIs) come later. When you want one, ask your assistant to set it up.

2. Get the class folder

- Get the class folder (**ai-coding-for-vfx-class**): clone it from GitHub, or use GitHub's **Code** → **Download ZIP**. Put it on **your own drive**, e.g. `Documents`. If it arrives as a .zip, unzip it first.
- **Not** inside Google Drive, OneDrive or Dropbox. Synced folders can make video jobs crawl and confuse the tools. Copy footage *out* to a local folder to work on it.
- **Open this folder in your app**. The assistant can only work inside the folder you give it, and this one comes with rules that make it your class tutor, with the studio safety rules and skills already loaded.
- The folder contains a few **hidden items** (`.claude`, `.agents`, `.gitignore`). A Mac hides anything starting with a dot; that's normal. Don't delete them: they hold the skills.

Mac only: the first time the app reads a folder, macOS may ask for permission ("...would like to access files in your Documents folder"). Click **Allow**. If you clicked Don't Allow, fix it in System Settings → Privacy & Security → Files and Folders.

3. Check your machine

Paste **setup prompt 0**. It only looks, changes nothing, and tells you which of the next steps you still need.

4. ffmpeg

The free tool that does all the video conversion. Paste **setup prompt 1**.

Manual:

System	Command
Windows	<code>winget install --id Gyan.FFmpeg -e</code>
Mac	<code>brew install ffmpeg</code> (install Homebrew first, below)
Ubuntu / Debian	<code>sudo apt install ffmpeg</code>
Fedora	enable RPM Fusion, then <code>sudo dnf install ffmpeg</code>

Check it: `ffmpeg -version`. For HDR phone footage you need a build with **zscale**; the ones above have it.

Installing Homebrew on a Mac (the Mac's package manager, needed once):

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Then follow the "Next steps" it prints to add brew to your PATH.

5. Python + uv

uv installs Python for you and keeps each tool's add-ons separate, so nothing breaks. Paste **setup prompt 2**.

Manual:

System	Command
Windows	<code>winget install --id astral-sh.uv -e</code>
Mac	<code>brew install uv</code>
Linux	the one-line installer below

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Check it: `uv --version`.

6. Git + GitHub

Git keeps every version of your code (your undo button); GitHub stores it online, **privately**.

1. Make a free account at **github.com** if you don't have one.
2. In **your own project folder** (not the class folder, which is John's repo), paste **setup prompt 3**. It installs git and the GitHub CLI, signs you in (you'll confirm a code in your browser), turns your folder into a **private** repo with the starter-pack `.gitignore`, and pushes your first commit.

Manual:

System	Command
Windows	<code>winget install --id Git.Git -e</code> and <code>winget install --id GitHub.cli -e</code>
Mac	<code>brew install git gh</code>
Ubuntu / Debian	<code>sudo apt install git gh</code>

Then sign in with `gh auth login` and follow the prompts.

7. Starter pack for your own projects

- The class folder is for **learning and practice**. Real show work goes in its own folder.
- For each real project folder, paste **setup prompt 4** with the path to the class folder's `starter-pack/`. It copies the studio rules, skills, prompts and `.gitignore` in, without overwriting anything.
- **Edit the "Naming" and "Delivery defaults" sections** of the copied `AGENTS.md` to match your show. Or ask: *"Update the naming section in AGENTS.md to use SHOW_SEQ_SHOT_TASK_v###."*

8. Optional extras

Extra	Prompt	What it gives you
MCP for Blender	5	The assistant can drive Blender directly: build scenes, set lights, render. (Needs uv.) Project page: github.com/ahujasid/mcp-for-blender .
Node.js	6	Needed for web apps and Tauri. Manual: <code>winget install --id OpenJS.NodeJS.LTS -e</code> · <code>brew install node</code> · Linux: use your distro's <code>nodejs</code> or <code>nvm</code> .

Extra	Prompt	What it gives you
Tauri	7	Small, fast desktop apps. Big first-time install (Rust and build tools) and a slow first build; that's normal.

Troubleshooting

"Command not found" right after installing. Quit the app completely and reopen it. New tools are only picked up by apps started after the install. On Windows, a restart sometimes helps.

It can't see or write to my folder. Check you opened the right folder in the app. On a Mac, check System Settings → Privacy & Security → Files and Folders.

Everything is very slow. Is the folder in Google Drive, OneDrive or Dropbox? Copy the files to a local folder. Cloud "placeholder" files download one at a time.

Antivirus or SmartScreen warns about something. Only continue if it's something you asked for from an official source (the sites in this guide). If unsure, ask your assistant what it is first, then ask John.

The assistant keeps failing at an install. Paste the "If something goes wrong" prompt with the error, or start a fresh session and try again. Still stuck? Use the manual commands above. Only then ask John, and tell him what you and your assistant already tried.

Setup prompts

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Paste one of these into Claude Code, Codex or Antigravity (desktop app is fine). The assistant works out whether you're on Windows, Mac or Linux and does the installing for you. **It will ask before installing anything. Read what it says, then approve.**

Do them in order the first time. Each one is safe to re-run.

0. Check my machine (read-only, nothing changes)

```
Check my computer and tell me, in plain language:
- which operating system and version I'm on
- whether these are installed, and which version: git, GitHub CLI (gh),
  Python, uv, ffmpeg (and whether it has zscale), Node.js, Blender
Don't install or change anything. Just give me a short table and tell me which
of the setup steps I still need.
```

1. Install ffmpeg

```
Install ffmpeg on this computer so I can convert video.
Use the normal package manager for my system (winget on Windows, Homebrew on
Mac, apt or dnf on Linux). I need a full build that includes zscale for HDR
tone-mapping.
Ask me before installing anything, explain each step in one line, and tell me
how to uninstall it. When you're done, prove it works by running
ffmpeg -version and checking zscale is listed.
```

2. Python and uv

```
Set me up to run small Python tools: install uv (it manages Python for me).
Ask before installing, explain each step in one line, and tell me how to
uninstall. Then prove it works by running a one-line Python "hello" with uv.
```

3. Git, GitHub and my first private repo

Help me set up GitHub:

1. Install git and the GitHub CLI (gh) if they're missing.
2. Sign me in to GitHub with gh (I'll do the browser step when you tell me).
3. Turn this folder into a PRIVATE GitHub repo. Add the .gitignore from the starter pack so media, caches and keys are never committed.
4. Make the first commit and push it.

Explain each step in one line, as if I've never used git. Ask before installing anything. At the end, show me the repo's URL and confirm it's private.

4. Install the starter pack into a project folder

I have the class starter pack at: <PASTE THE STARTER-PACK FOLDER PATH>

Copy these into this folder, without overwriting anything that's already here: AGENTS.md, CLAUDE.md, .gitignore, the .claude/skills folder, the .agents/skills folder and the prompts folder.

Show me the list of what you'll copy first (dry run), then do it when I say yes.

Then read AGENTS.md and tell me in 3 lines what rules you'll follow.

5. MCP for Blender (formerly Blender-MCP)

Set up "MCP for Blender" (github.com/ahujasid/mcp-for-blender) so you can control Blender for me.

- Check the project's current README for the install steps for THIS tool (Claude Code, Codex or Antigravity) and my operating system.
- It needs uv, and a Blender add-on that I install inside Blender. Walk me through the Blender part step by step.
- Turn on its safe mode if the README offers one.
- Ask before installing anything. When it's done, prove it works by asking Blender for the names of the objects in the current scene.

6. Node.js (for web apps and Tauri)

Install Node.js LTS so I can build small web apps. Use the normal package manager for my system, ask before installing, and prove it works with `node --version` and `npm --version`.

7. Tauri (desktop apps)

Set my computer up to build a Tauri desktop app.
Check the current Tauri prerequisites page for my operating system (Rust, plus Windows: Microsoft C++ Build Tools and WebView2; Mac: Xcode Command Line Tools; Linux: the listed system packages).
Tell me what's missing and how big the downloads are, ask before installing, and warn me that the first build is slow. Prove it works by creating and running the Tauri starter template with React, Vite and TypeScript.

If something goes wrong

That didn't work. Here's the error: <PASTE IT>
Explain what went wrong in one plain sentence, then fix it. Don't try more than two approaches without checking with me.